Exercise on Equation Solutions

- How many solutions does the equation
$$x_1 + x_2 + x_3 = 5$$
have, where x_1, x_2, and x_3 are nonnegative integers?
- Equivalent problem: Suppose that a cookie shop has 3 different kinds of cookies. In how many different ways can 5 cookies be chosen?
- **Solution**
- Each solution corresponds to a way to select 5 items with repetition from 3 element types; x_1 items of type 1, x_2 of type 2, and x_3 of type 3.
- Equivalently: I wish to select 5 indistinguishable objects from 3 distinguishable boxes x_1, x_2, and x_3.
- By the previous theorem, it follows that there are
$$C(3 + 5 - 1, 5) = C(7, 5) = \frac{7!}{5!\,2!} = 21$$
solutions.